

2023 AMC 10A Problem 6 - Solution

Review the full statement and step-by-step solution for 2023 AMC 10A Problem 6. Great practice for AMC 10, AMC 12, AIME, and other math contests

2023 AMC 10A Problem 6

Problem:

An integer is assigned to each vertex of a cube. The value of an edge is defined to be the sum of the values of the two vertices it touches, and the value of a face is defined to be the sum of the values of the four edges surrounding it. The value of the cube is defined as the sum of the values of its six faces. Suppose the sum of the integers assigned to the vertices is 21. What is the value of the cube?

Answer Choices:

- A. 42
- B. 63
- C. 84
- D. 126
- E. 252

Solution:

Each vertex of the cube touches 3 edges, so the sum of the values of the 12 edges is 3 times the sum of the values of the 8 vertices. Each edge touches 2 faces, so the sum of the values of the faces is twice the sum of the values of the edges. Thus the value of the cube is $2 \cdot 3 \cdot 21 = (D) \boxed{126}$.

For example, assigning 21 to one vertex and 0 to the other seven vertices will result in the given values. Specifically, there will be three edges with value 21 and nine edges with value 0. This means that there will be three faces with value 42 and three faces with value 0.

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